

Project Rubric

EE 451 (Parallel and Distributed Computing)

Spring 2026

- Grading breakdown for the course project
 - Project Proposal: 25%
 - Project presentation: 25%
 - Project final report: 50%

This rubric describes the expectations and deliverables for the project. As per the syllabus (page 23), the project proposal is worth 25% of the project grade, the presentation is worth 25% and the final report is worth 50%. Overall, the project is worth 20% of the total course grade.

Be sure to update the tracking spreadsheet for what ideas students have and team names here: <https://docs.google.com/spreadsheets/d/1nfmzaTuKuFggxtHh1EsfAjf8lhELz31xne467N8X4FQ/edit?usp=sharing>

Project Proposal:

First Draft: Due on Wednesday, March 11, @ 11:59PM. [5% of project grade]

- One page
- Team members clearly listed
- High-level goals
- Project category identified:
 - Research, Implement, Evaluate
 - Implement and Evaluate
 - Full Application Parallelization

Final Draft: Due on Tuesday, March 24 @ 11:59PM: 20%

Project Proposal Rubric [out of 100, scaled to 20%]

Section of the Proposal	Requirements
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Introduction [20 points]	• Must include the background on the problem (i.e. what you are accelerating, why is it non-trivial to accelerate). • Contributions for the project (i.e. what you plan to have as a deliverable for the project, what your work aims to do). • Hypotheses of the project (i.e. what you think will happen based on your intuition from this course).
Description [60 points]	• Parallelization techniques if proposing a new technique • Selection criteria of benchmarks if comparing parallel algorithms • Description of the application if developing an application • Implementation platform
Evaluation [20 points]	• Mention any datasets, evaluation metrics you will be using • Mention what deliverables you will have code-wise

Project Presentation Rubric [out of 100%, scaled to 25%]

Category	Points	Excellent (Full Credit)	Partial Credit	Poor / Missing
Problem & Motivation	10	Clear problem definition; strong motivation; non-trivial parallel challenge identified	Problem stated but weak motivation or unclear challenge	Problem unclear or trivial
Technical Background	10	Demonstrates understanding of relevant parallel concepts (memory-bound, scalability, communication, etc.)	Some concepts mentioned but not well connected	Little or no technical grounding
Parallelization Strategy	20	Clear decomposition; addresses data dependencies, load balancing, synchronization	Strategy described but missing important considerations	Vague or incorrect parallel approach
Implementation Details	10	Platform clearly explained (Pthreads, GPU, MPI, etc.); architecture/diagram included	Some implementation details missing	Platform unclear or poorly described
Experimental Setup	10	Hardware specs, datasets, baseline clearly defined	Missing some details	No clear setup description

Results & Figures	15	Clear graphs (execution time, speedup, scalability); labeled axes; readable	Graphs present but unclear or incomplete	Minimal or no quantitative results
Analysis & Interpretation	15	Explains bottlenecks, scalability limits, anomalies; connects to theory (Amdahl's Law, memory bandwidth, etc.)	Some explanation but shallow analysis	No meaningful analysis
Conclusion & Hypothesis Validation	5	Clearly states whether hypothesis held and why	Vague conclusion	No conclusion
Clarity & Organization	5	Logical flow; slides readable; clear transitions	Some organization issues	Disorganized presentation
Team Participation & Delivery	10	Balanced participation; confident delivery; within time	Uneven participation	One person presents or poor delivery

Project Report Rubric [out of 100, scaled to 50%]

Category	Points	Excellent (Full Credit)	Partial Credit	Poor / Missing
Introduction & Motivation	10	Clear background; non-trivial problem; well-defined contributions	Motivation present but weak or vague	No clear problem statement
Related Work / Context	5	Positions project within existing work	Minimal reference to related work	No references
Technical Approach	20	Clear algorithm description; diagrams/pseudocode; complexity discussion; correct parallel reasoning	Implementation described but lacking clarity or depth	Incomplete or incorrect technical explanation
Parallelization Design	10	Addresses data dependencies, load balancing, synchronization, communication costs	Partial discussion of design issues	No discussion of parallel challenges

Implementation Details	10	Platform, architecture, parameters clearly documented	Some details missing	Cannot reproduce implementation
Experimental Setup	10	Hardware specs; dataset description; baseline defined; reproducible setup	Some details missing	Experimental setup unclear
Results (Quantitative)	10	Execution time, speedup, scalability plots included and clear	Some results missing or poorly presented	Minimal or no quantitative results
Analysis & Discussion	15	Deep analysis of bottlenecks; scalability limits; memory vs compute bound; Amdahl's Law; unexpected behavior explained	Basic explanation without depth	No real analysis
Conclusion & Future Work	5	Reflects on hypothesis; discusses limitations and improvements	Weak conclusion	Missing conclusion
Writing Quality & Organization	5	Professional, clear structure, figures referenced properly	Some clarity issues	Poor writing and organization

Extra Credit:

If your work is evaluated to be publishable in a strong venue (IEEE, etc.) and ready to submit before the last day to finalize the grades, then there will be a major extra credit part.

Depending on the quality of work (evaluated by the instructor, the TAs and possibly Dr. Prasanna), the extra credit could be between 20% and 30% extra. This is a challenging extra credit part to work on.

If your work includes meaningful hardware components (GPUs, FPGAs, etc.) there would be an extra credit part 1% to 5%.

We also evaluate the quality of your work relative to the rest of the teams and assign extra credit points of 1% to 5% for the top 3 teams.

